Why substitution costs so much in `seg/model`, and a proposal: irrelevant equality proofs in sets

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1 The feature: El along a code map

The base category is generic. Let D be a context and $c : D \to \mathcal{U}$ a map into the universe: codes $c(x) \in \mathcal{U}(I)$ for $x \in D(I)$, with naturality proofs

$$n_c(f, x) : c(x \cdot f) = c(x) \cdot f \qquad (f : J \to I).$$

The type El_c over D has fibres $\mathrm{El}_c(x) = \mathrm{El}(c(x))$. For a map of contexts $s : D' \to D$ (with naturality proofs n_s) we want

$$\mathrm{El}_c[s] \;=\; \mathrm{El}_{c \circ s}. \qquad (*)$$

2 What happens now

Types act along proofs. A type A over G consists of fibres and an action $A(x) \to A(y)$ for $f : J \to I$ and *a proof* $e : x \cdot f = y$ (`Presheaf.PTy`, field `tact`). The action of El_c is: act in El from $c(x)$ to $c(x) \cdot f$, then *transport* the result along a proof of $c(x) \cdot f = c(y)$, namely

$$p_c(e) \;=\; n_c(f, x)^{-1} \cdot c(e).$$

Transport along a proof is an operation on terms: different proofs give, a priori, different terms.

The two sides of $(*)$. Both sides have the same fibres, $\mathrm{El}(c(s(x)))$. Their actions differ only in the proof transported along:

$$\begin{aligned} \mathrm{El}_c[s] :&\quad p_c\big(n_s(f, x)^{-1} \cdot s(e)\big) \\ \mathrm{El}_{c \circ s} :&\quad p_{c \circ s}(e) \;=\; \big(c(n_s(f, x)) \cdot n_c(f, s(x))\big)^{-1} \cdot c(s(e)) \end{aligned}$$

These are two proofs of the *same* equation $c(s(x)) \cdot f = c(s(y))$ in $\mathcal{U}(J)$. Since $\mathcal{U}(J)$ is a set, they are equal, but only by a proof (using that $\mathcal{U}(J)$ is a set), not by computation. So $(*)$ is a theorem, `elSub`, and not a definitional equality: in the checker `refl` fails on $(*)$ but succeeds on its fibre part.

The cost. Every structure we put on El_c (its compositions, its fillers, the classifier, equivalence data) must then be moved along $(*)$ and shown to agree with the structure on $\mathrm{El}_{c \circ s}$. For the compositions this is `GElC.ElF.FibSub` (60 lines), which is in turn an obligation of the univalence operation (`tautCl-sub`). The same pattern, "two constructions agree up to proofs, so transport the structure and prove coherence", accounts for a large part of the development: of about 9800 lines, 39% prove that substitution commutes with a construction and 15% transport structures along equations of types.

3 The proposal

The rule. Make any two proofs of an equation between elements of a set *definitionally* equal:

$$\frac{p,\ q : \mathrm{Id}_A(a,b) \qquad A \text{ a set}}{p \equiv q}$$

In `seg` the sets we use are all of the form $\mathrm{Ix}(v)$ for a code v (fibres, codes, maps of the base), so the rule can be stated syntactically: two terms whose type is $\mathrm{Id}_{\mathrm{Ix}(v)}(a,b)$ are convertible. This is a rule of the checker (a kernel change), comparable to a sort of definitionally irrelevant propositions. Its consistency with the rest of the metatheory (univalence of the ambient universe) and the decidability of conversion with it have to be checked; we do not claim them here.

What it gives for $(*)$. The two proofs above are then the same, the two actions are the same term, and $(*)$ holds by definition. Consequently the substituted structure is *literally* the structure of El_{cos}: nothing is transported, and `elSub`, `FibSub`, and the obligation `tautCl-sub` become `refl`. The statements of the development do not change, and the base stays generic: the rule only concerns proofs, never the maps of the base.

What it does not give. Equations between *data* remain theorems. Examples: El along the Yoneda code of F is F (`elYon`) uses $g \circ \mathrm{id} = g$, an equation between maps of the generic base; the codes of Ext at a full point use full = true (`extOb-B`); a restricted cofibration is the substituted one only up to $\mathrm{pre}(g, \mathrm{pre}(h, Q)) = \mathrm{pre}(hg, Q)$. Transports along these stay. The rule removes exactly the transports whose only cause is that two *proofs* differ.

Remark. This is the analogue of Ehrhard's presentation of categories with families (`cwf2.pdf`) compared with Dybjer's: there, the lifted substitution $\sigma^+ = (\sigma\mathrm{p}, \mathrm{q})$ forces q to be transported along $A\sigma\mathrm{p} = A(\sigma\mathrm{p})$, and taking σ^+ primitive removes these transports while keeping the equations between types. Here the rule removes the transports along *proofs*, while the equations themselves stay. The moves between universe levels (8% of the development) are a separate issue and are not affected.